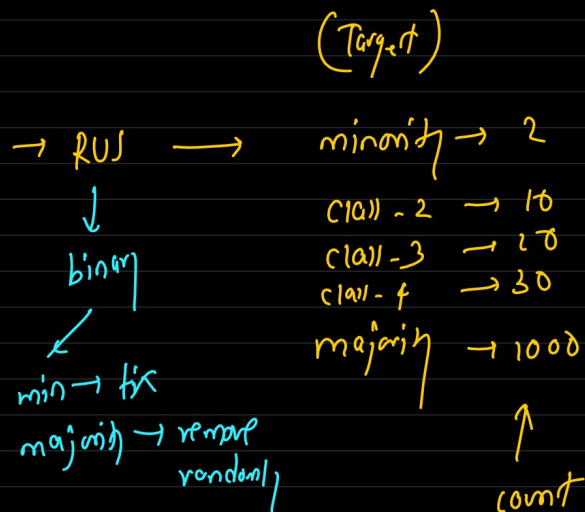


16-May-2026

Agenda:

FE-9

EDA-1



all → delete row from every single class in your dataset until its count matches the minority.

Apple → 100 → 30

Banana → 50 → 30

orange → 30 → 30

not majority → leave the majority class completely untouched. It look at every other class & delete row from them until they match the size of minority.

Apple → 100 → 100

Banana → 50 → 30

orange → 30 → 30

majority → It target only the single largest class & drop its rows until its count matches the minority

Apple → 100 → 30

Banana → 50 → 50

orange → 30 → 30

ROS → 'not majority' (increase count of minority)

→ find the biggest class & make it as the base.

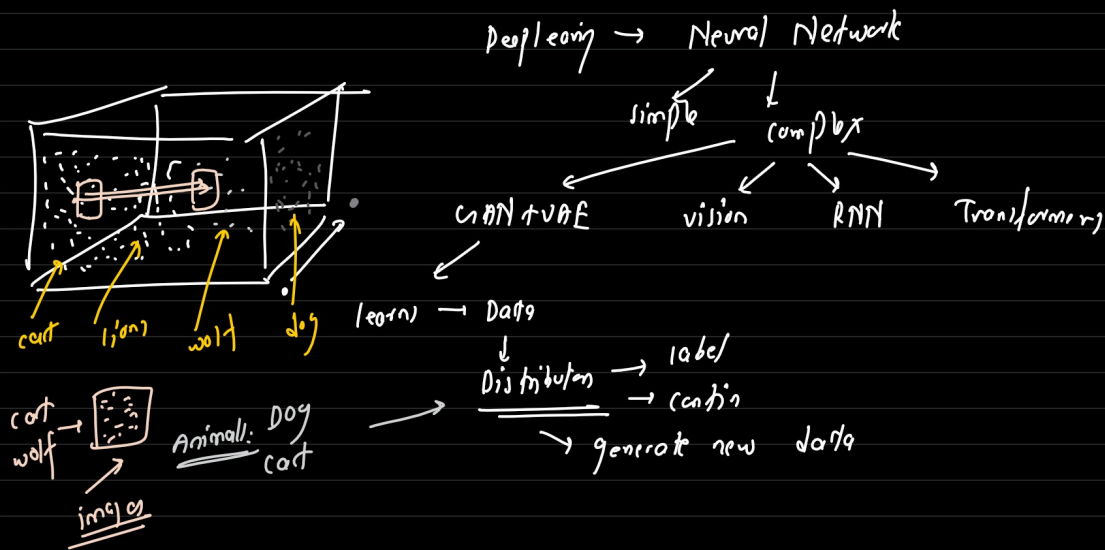
Apple → 100 → 100

Banana → 50 → 100

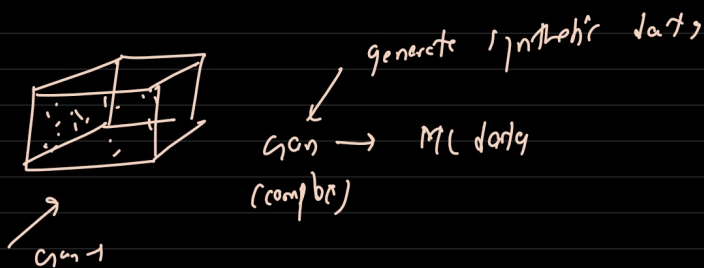
orange → 30 → 100

Completed - FE

Data imbalance $\rightarrow$ start $\rightarrow$ Handle imbalance



ML $\rightarrow$ classify $\rightarrow$ predict label (0, 1, 2)
 $\rightarrow$ regression $\rightarrow$ predict continuous (0.1, 0.2, 100.5...)



(EDA)

Exploratory Data Analysis

Data $\rightarrow$ Data Extraction $\rightarrow$ EDA $\rightarrow$ FE $\rightarrow$ ML
(source)

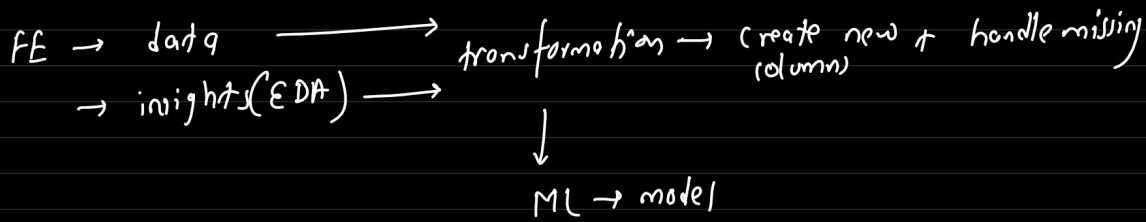
EDA $\rightarrow$ data $\rightarrow$ understand its patterns

$\downarrow$

$\rightarrow$ find mistakes

$\rightarrow$ discover relationship $\rightarrow$  EDA

$\rightarrow$ we do not change, just analyze it.

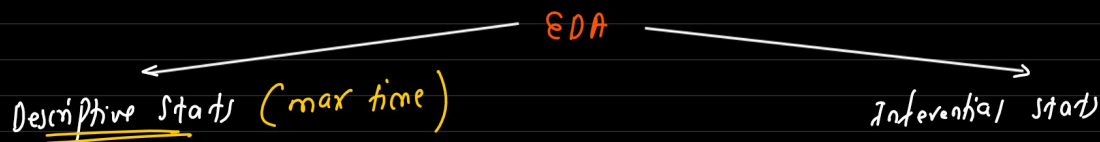
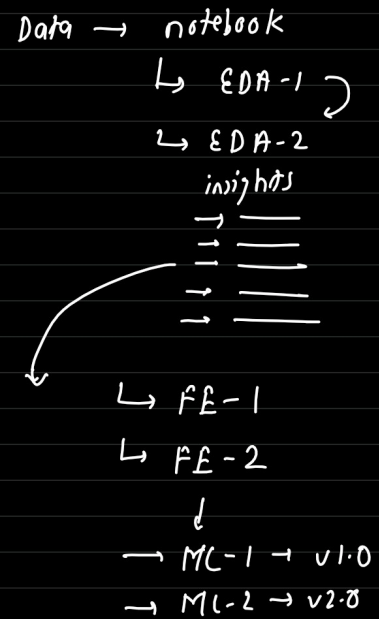


4 main tasks performed during EDA:

- checking data distribution
- Investigating Missingness pattern
- analyzing correlation & relationships
- spotting outliers

↑
starts

↑
visualization



Summarize & visualize what the data looks like

- Central tendency (mean, median, mode)
- Variability (Variance, STD, IQR)
- Frequency plots, histogram, boxplots ...
- Missing values, Outliers

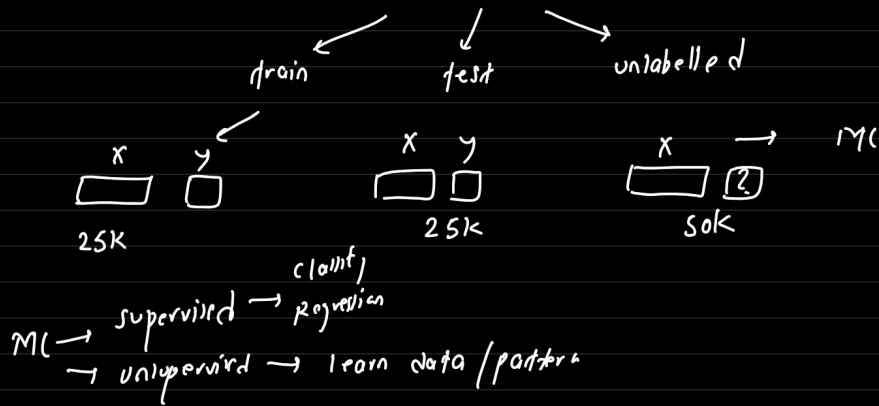
Ask : What is happening in the dataset?

Make inferences or generalization about the population from the sample

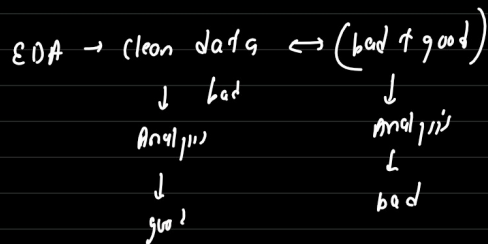
- Hypothesis test
- Confidence interval

Ask : Can we claim this effect exist in the larger population or is it just by chance?

Large Movie Review Dataset



- % of pos / -ve sentiment ✓
- missing values ✓
- distribution ✓
- duplicate rows ✓



NLP (specialization) → language

nltk → nlp task → english
 (library) → french

corpus → collection of text / data

stop-words: a, an, the, she, it, on, at, he

(common-words: which doesn't add any significant meaning)

n-grams:

Sequence of n words that appear together in a text

n=1, Unigram: movie, good, bad, hello, data

n=2, Bigram: great movie, very good, not good, hello person

n=3, Trigram: waste of time, best movie ever

← not good

useful

Analysis

↓
unigram → not, good → ntk
 ↓ ↓
 -ve +ve

ntk

→ Review → row(0) →

positive = 0

negative = 0

for i in words:

if i is +ve:

positive += 1

if i is -ve:

negative += 1

if positive > negative:
positive

else:
negative

bigrams:

not good → word

movie not_good (together, words mean something)
↓ ↓
0 -ve

Trigram:

not good decisions